

Robot-Agnostic and Adaptive PBVS with Auto-Calibration

I. Result and Discussion

A. Simulation Results:

Case No	Case Name	System Name	Trial	IP	TP	ET	λ_1	λ_2	λ_3	λ_4	λ_5	λ_6
2	Manipulator	elephant	T1	(0.032, -0.06, 0.43)	(0.19, -0.10, 0.26)	(0.03,0.01,0.03)	0.1	0.012	0.02	0.02	0.01	0.03
3	NHMR	PioneerP3dx	T1	(0.0, 0.0)	(-4.0, -1.2)	(0.01,0.01)	0.28	0.19	-	-	-	0.1
			T2	(0.0, 0.0)	(-8.396, -0.763)	(0.01,0.01)	0.22	0.22	-	-	-	0.22
4	Hybrid	Pioneer_P3dx	T1	(-1.0, 0.0)	(-6.284, -0.409)	(0.01,0.01)	0.22	0.22	-	-	-	0.22
		Parrot_Drone	T1	(0.0, 0.0, 0.9)	(NHMR, 1.15)	(0.09,0.11,-0.05)	0.5	0.1	0.001	0.5	0.1	0.001

TABLE I: Parameter values of the proposed approach. TP = Target Position (m), IP = Initial Position (m), ET = Error Threshold (m), Trial-1 = T1 Trial-2 = T2, Trail-3 = T3. Hybrid = NHMR_with_Drone. Here λ_1 – λ_6 denote the proportional gains .

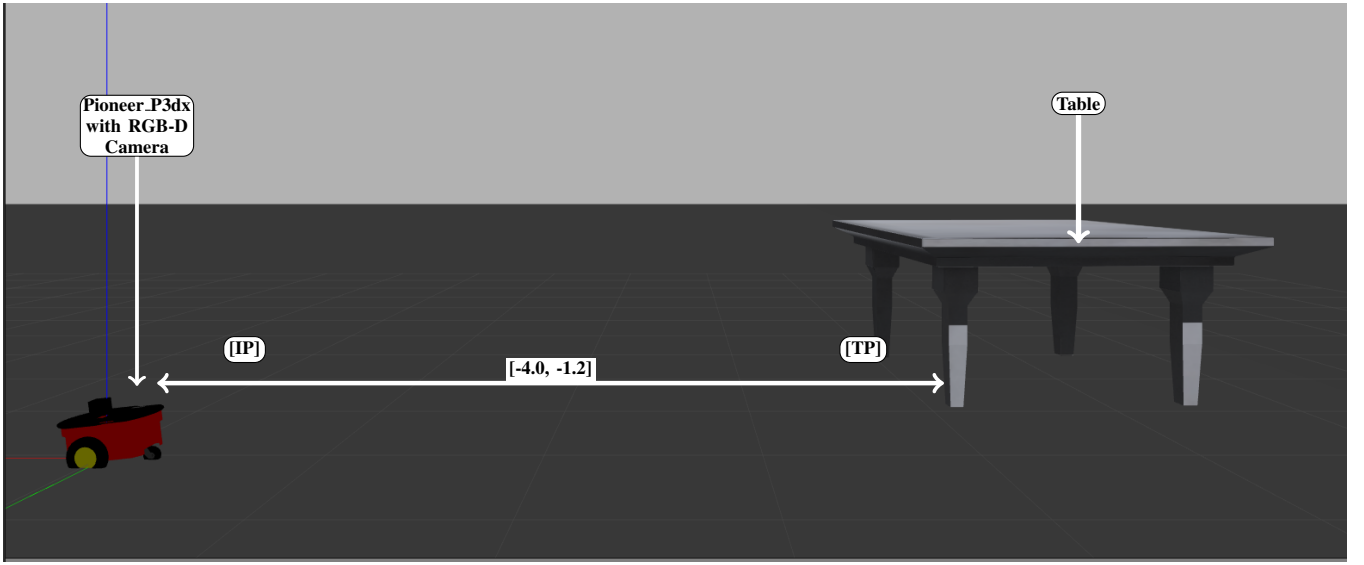


Fig. 1: Experimental setup for NHMR to reach the target table for T1.

1) **Case 1 - NHMR:** To further validate the proposed approach, two experimental trials (T1 and T2) were conducted using the P3dx mobile robot platform. The second and third rows of Table I detail the experimental configuration for T1 and T2, respectively, including the initial position, goal position, and camera parameter settings, ensuring non-trivial coupled displacement along both the X and Y axes during navigation. In both trials, a table object was used as the target.

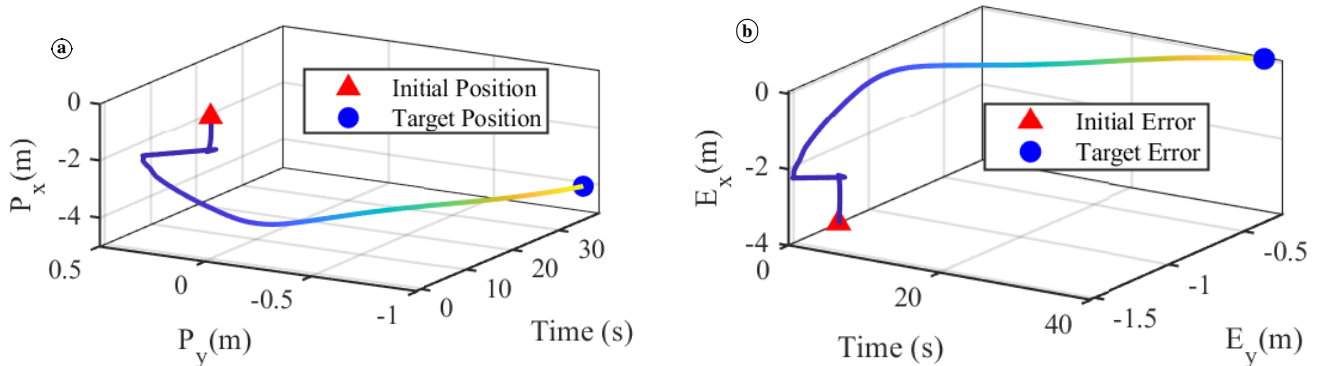


Fig. 2: Case 3 shows the (a): NHMR Position Path((P_x, P_y) vs Time (b): NHMR Error Path((E_x, E_y) vs Time for experimental setup of T1.

T1: The experimental setup for T1, depicted in Fig. 1, considers an target position with moderate displacements along both the x- and y-axes. The resulting position path and error graph are shown in Fig.2 , demonstrating that the proposed mathematical law with p control well drives the mobile robot to the target.

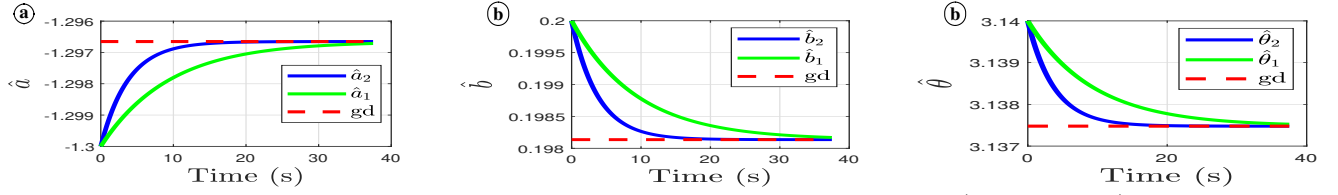


Fig. 3: Case 3: Comparative results of adaptive vs non-adaptive w.r.t the ground truth (GT) for (a) $\hat{a}$, (b) $\hat{b}$, (c) shows the $\hat{\theta}_y$ for the NHMR (T1).

The camera parameter estimation results for the NHMR are presented in Fig. 3, where the adaptive estimates ($\hat{a}_1$, $\hat{b}_1$, and $\hat{\theta}_{y1}$) and the non-adaptive estimates ($\hat{a}_2$, $\hat{b}_2$, and $\hat{\theta}_{y2}$) are evaluated against the corresponding ground-truth camera parameters. Although both estimators asymptotically converge to the true parameter values, the adaptive estimator exhibits a significantly faster convergence rate with reduced transient estimation error, whereas the non-adaptive estimator converges more slowly to the true parameter value.

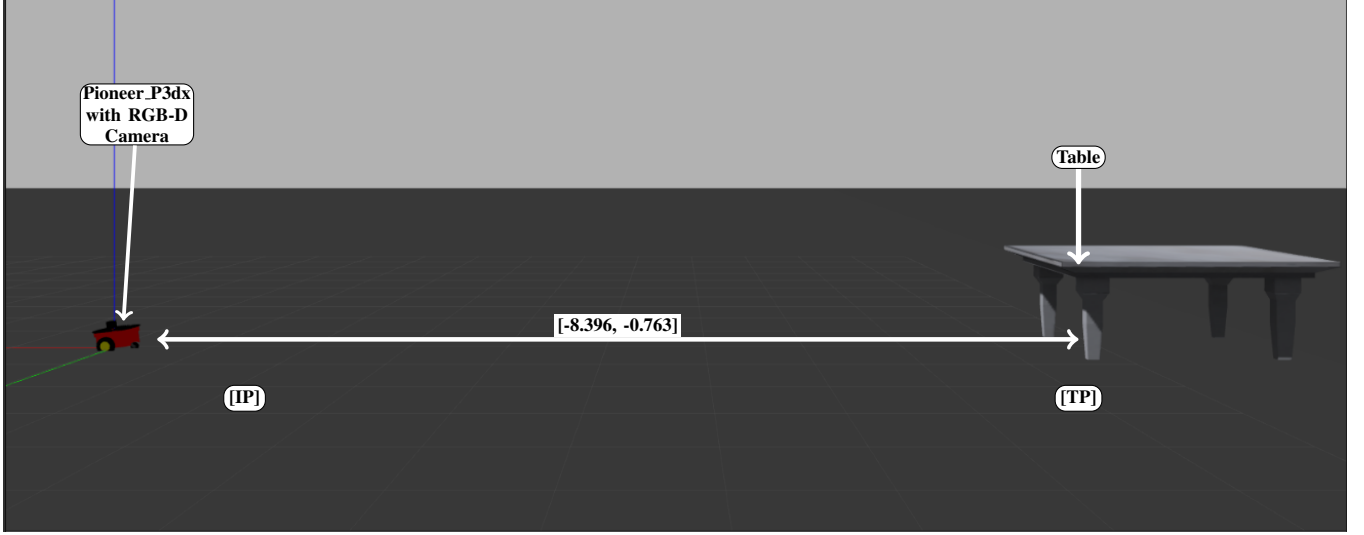


Fig. 4: Experimental setup for NHMR to reach the target table for T2.

T2: The experimental setup for T2, depicted in Fig. 4, considers an target position with a significantly larger displacement along the x-axis and a comparatively smaller displacement along the y-axis.

The resulting position path and error graph are shown in Fig. 5, demonstrating that the proposed mathematical law with p control well drives the robot to the target with significant motion along both axes. The camera parameter estimation results for the NHMR are presented in Fig. 6, where the adaptive estimates ($\hat{a}_1$, $\hat{b}_1$, and $\hat{\theta}_{y1}$) and the non-adaptive estimates ($\hat{a}_2$, $\hat{b}_2$, and $\hat{\theta}_{y2}$) are evaluated against the corresponding ground-truth camera parameters. Although both estimators asymptotically converge to the true parameter values, the adaptive estimator exhibits a significantly faster convergence rate with reduced transient estimation error, whereas the non-adaptive estimator converges more slowly to the true parameter value.

The auto-extrinsic calibration results for both trials are summarized in Table II, where the first three rows report the estimated camera parameters $\hat{a}$, $\hat{b}$, and $\hat{\theta}_y$ for T1, and the last three rows correspond to T2. For each parameter, the initial value (IV) and ground-truth (GT) value are provided, along with the convergence time for both the adaptive estimator and the non-adaptive estimator (NAL). The results demonstrate that the adaptive estimator consistently achieves faster convergence across all parameters and trials compared to the non-adaptive estimator. The overall 2D RMSE, computed was 0.4 m for T1 and 0.12 m for T2. The consistent tracking performance and convergence of the adaptive camera parameters across two distinct configurations demonstrate that the proposed framework work well to different trails.

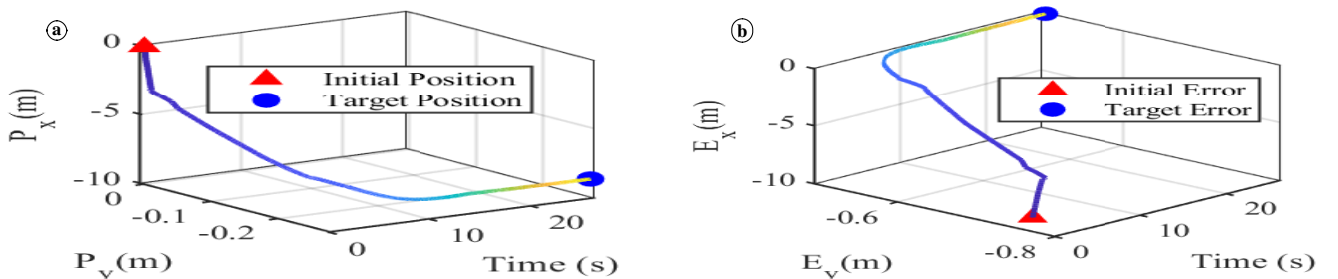


Fig. 5: Case 3 shows the (a): NHMR Position Path(P_x, P_y) vs Time (b): NHMR Error Path(E_x, E_y) vs Time for experiments of T2.

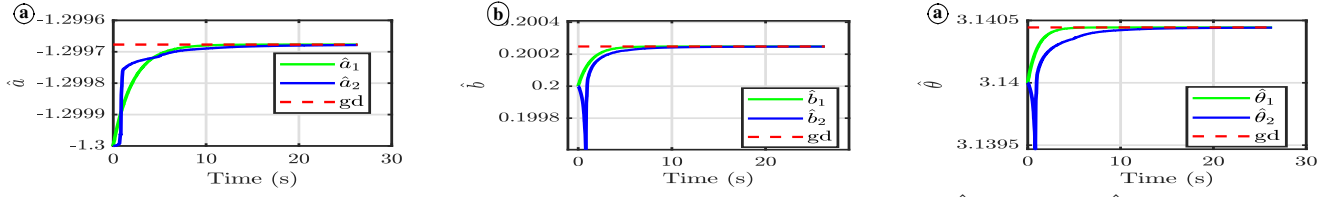


Fig. 6: Case 3: Comparative results of adaptive vs non-adaptive w.r.t the ground truth (GT) for (a) $\hat{a}$, (b) $\hat{b}$, (c) shows the $\hat{\theta}_y$ for the NHMR (T2) .

Trial	Parameter	IV	GD	AL(Time(s))	NAL(Time(s))
T1	$\hat{a}$	-1.3	-1.2966	13.97	36.81
T1	$\hat{b}$	0.2	0.19813	14.52	37.12
T1	$\hat{\theta}_y$	3.14	3.1373	14.57	37.27
T2	$\hat{a}$	-1.3	-1.2997	7.85	12.85
T2	$\hat{b}$	0.2	0.20025	7.40	23.71
T2	$\hat{\theta}_y$	3.14	3.14044	7.30	42.12

TABLE II: Case 3: Comparison between adaptive and non-adaptive estimator approaches for different NHMR trials.

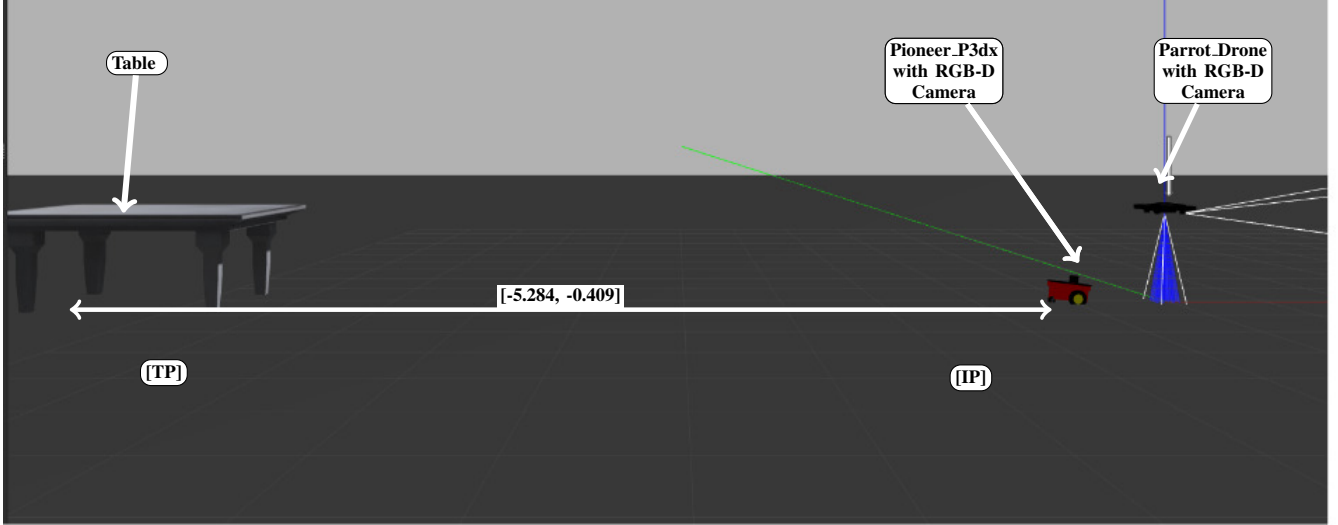


Fig. 7: Experimental setup for the P3dx NHMR reaching a table and parrot-drone following the NHMR.

2) **Case 4: Drone with NHMR:** In this case, we consider we used two target where one of the targets is static (table) and one target is dynamic (moving NHMR). The goal is to track the moving target (NHMR) using a drone. The moving NHMR will detect the table with its RGBD camera and move toward it. Our derived model, which minimizes the error to reach the target position and estimates the body-to-camera parameters using adaptive learning, all within a single loop. The experimental setup shown in Fig. 7 shows the drone reaching the table(object). Analyzing the result position path of both systems in Fig. 8, we observe that the proposed approach works successfully for the NHMR with drone (hybrid) case. Fig. 8 also shows the estimated body-to-camera translation parameter ($\hat{a}$) for the NHMR. The results in Fig. 9 show the estimated camera extrinsic parameters for the NHMR and the drone. Subfigures (a) and (b) present the estimated translation parameter ($\hat{b}$) and rotation parameter ($\hat{\theta}_y$) for the NHMR, respectively. Subfigures (c)–(h) present the adaptive (denoted by $\hat{a}_1$, $\hat{b}_1$, $\hat{c}_1$, $\hat{\theta}_{r1}$, $\hat{\theta}_{p1}$, and $\hat{\theta}_{y1}$) and non-adaptive (denoted by $\hat{a}_2$, $\hat{b}_2$, $\hat{c}_2$, $\hat{\theta}_{r2}$, $\hat{\theta}_{p2}$, and $\hat{\theta}_{y2}$) estimates of the drone camera extrinsic parameters over time. The auto-extrinsic calibration results have been summarized in Table III.

The overall 3D RMSE was found to be 1.35 m(drone), and 2D RMSE was found to be 0.21 m(NHMR).

Parameter	IV	GT	AL(Time(s))	NAL(Time(s))
$\hat{a}(NHMR)$	-1.3	-1.30002	5.06	20.27
$\hat{b}(NHMR)$	0.2	0.200049	0.92	3.77
$\hat{\theta}_y(NHMR)$	3.14	3.140002	1.57	2.07
$\hat{a}(drone)$	0.09	0.1299	5.06	20.27
$\hat{b}(drone)$	0.04	0.0499	14.43	25.87
$\hat{c}(drone)$	0.05	0.0498	24.07	26.08
$\hat{\theta}_r(drone)$	0.18	0.1795	21.89	24.82
$\hat{\theta}_p(drone)$	0.09	0.0909	24.8	29.70
$\hat{\theta}_y(drone)$	0.524	0.5238	20.27	25.80

TABLE III: Case 4: Comparison b/w adaptive and non-adaptive estimator approach.

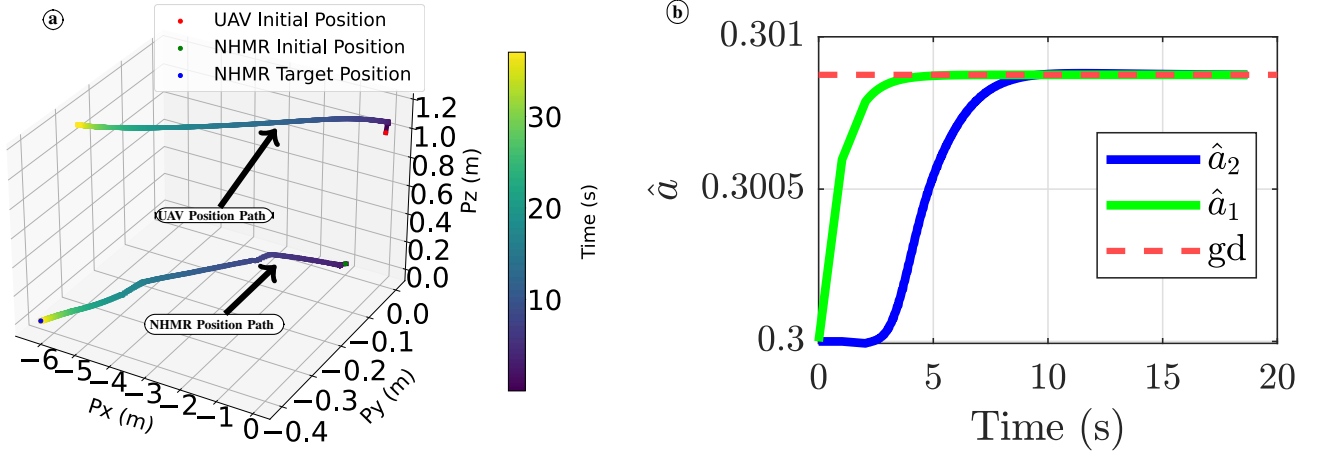


Fig. 8: Case 4: (a) shows the drone position path and NHMR position path (target) and (b) Comparative results of adaptive vs non-adaptive w.r.t the GT for $\hat{a}$ for the drone with NHMR experiments, Here UAV=Drone.

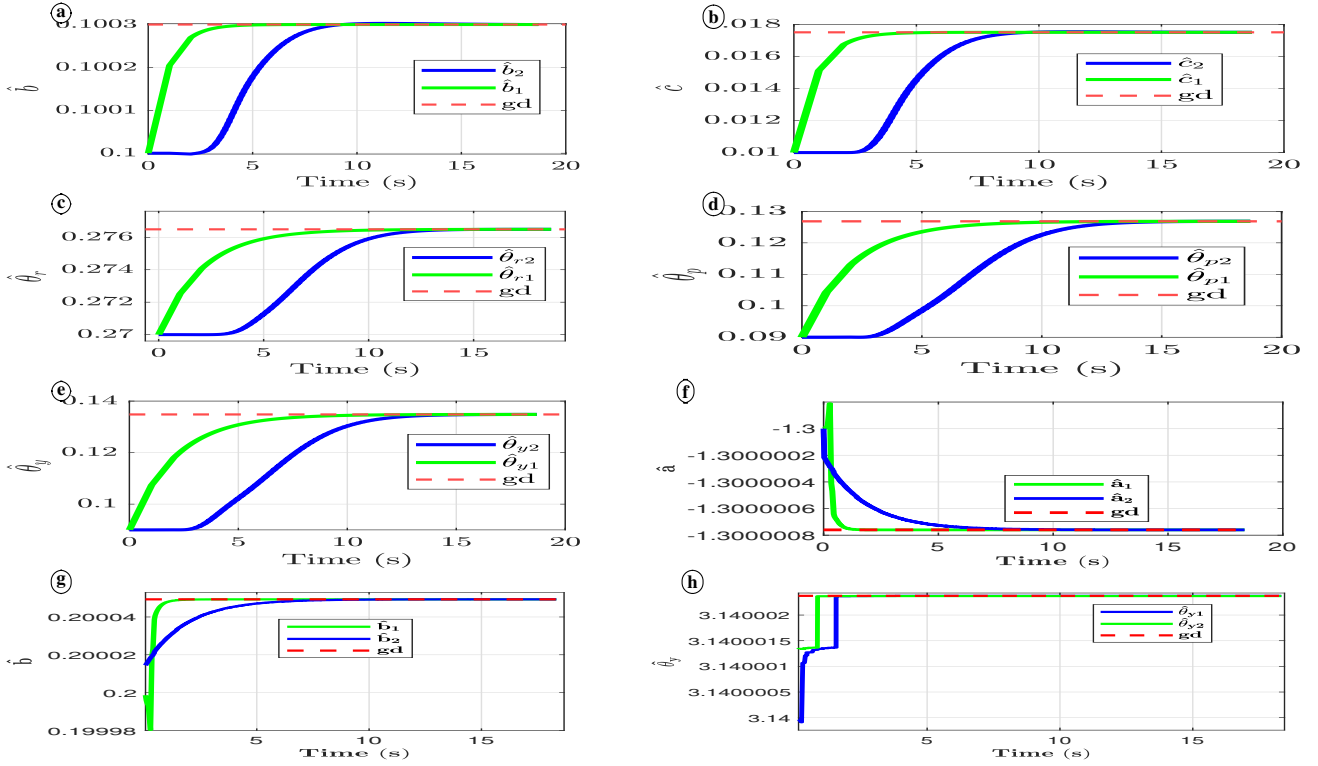


Fig. 9: Case 4: Comparative results of adaptive vs non-adaptive w.r.t the GT for (a) the drone system's $\hat{b}$, (b) the drone system's $\hat{c}$, (c) the drone system's $\hat{\theta}_p$, (d) the drone system's $\hat{\theta}_p$, (e) the drone system's $\hat{\theta}_y$, (f) the NHMR system's $\hat{a}$, (g) the NHMR system's $\hat{b}$, and (h) the NHMR system's $\hat{\theta}_y$ for the drone with NHMR experiments.

II. Real-Time Experiments

A. Case 2: Elephant Robotic Arm

To further validate the proposed approach, real-time elephant robotic manipulator. The first rows of Table I detail the experimental configuration and the experimental setup shown in Fig. 10 consists of a real Elephant 6-DOF robotic arm and an RGB camera equipped with an ArUco marker for Cartesian end-effector position measurement. The RGB camera detects the ArUco marker attached to the end-effector and provides the Cartesian position in the x , y , and z directions. The proposed approach generates the required Cartesian velocity for the end-effector motion. However, the Elephant robotic arm does not support direct Cartesian velocity commands. Therefore, the generated velocity command was converted into the corresponding position command and sent to the robotic arm to drive the end-effector toward the desired target position. Using an RGB camera, we calculated the ArUco marker target position. The target position was obtained as $X = 0.19$ m, $y = -0.10$ m, and $z = 0.26$ m. The experimental results for the real Elephant robotic arm are shown in position trajectory Fig. 11a and error trajectory Fig. 11b. The position trajectory demonstrates that the end-effector successfully moves from its

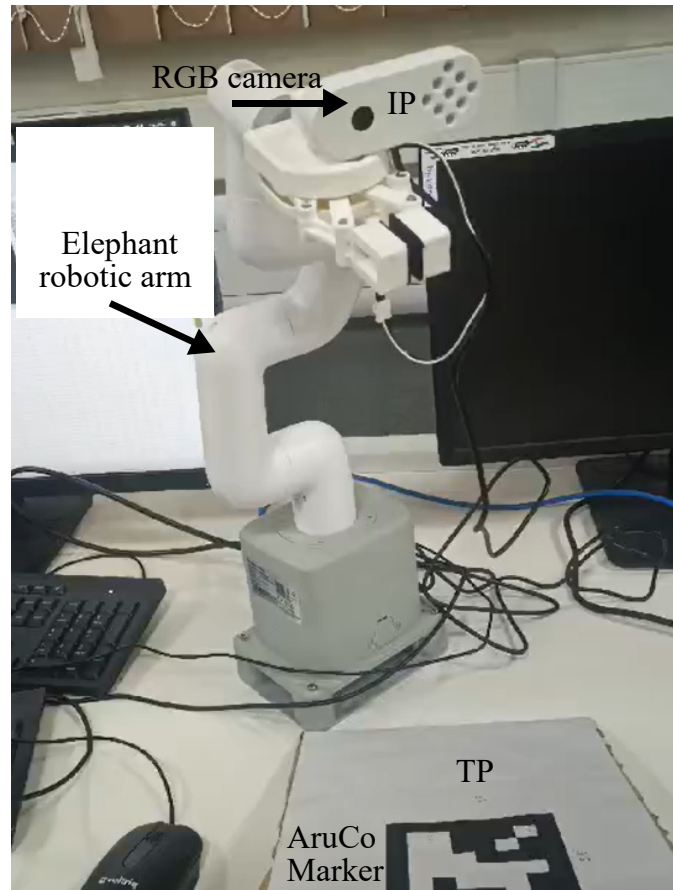


Fig. 10: Case 2: Experimental Set-Up for the real-elephant robotic arm with target aruco marker.

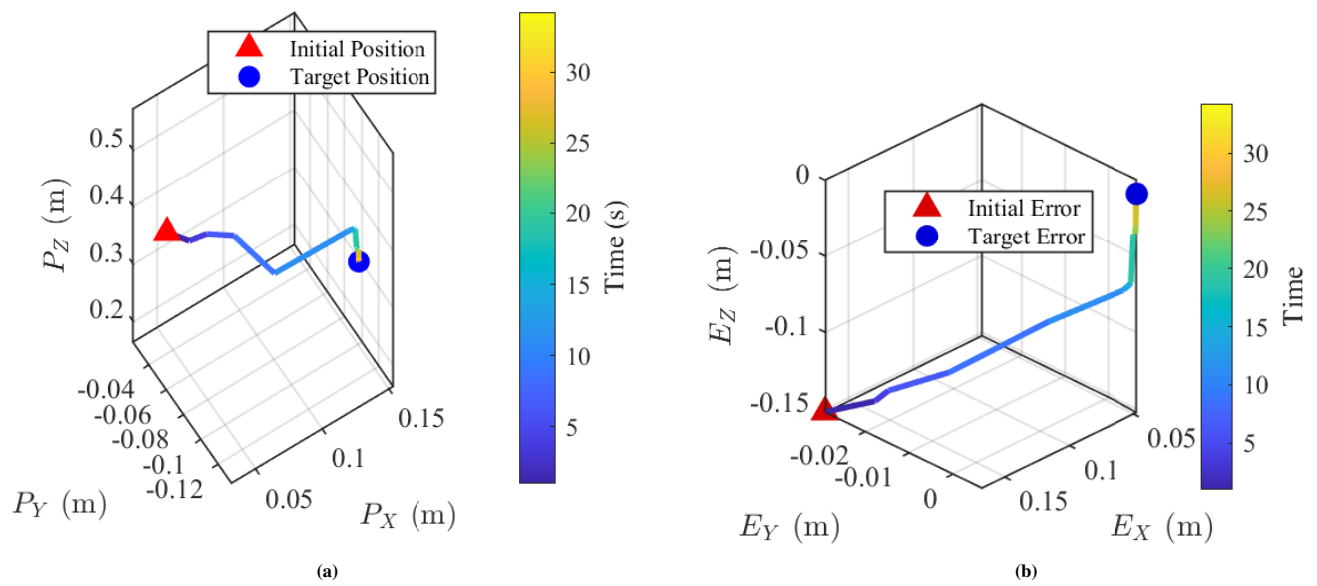


Fig. 11: Case 2: shows the (a) position path (P_x, P_y, P_z) and (b) error path (E_x, E_y, E_z) for the elephant robotic-manipulator experiment.

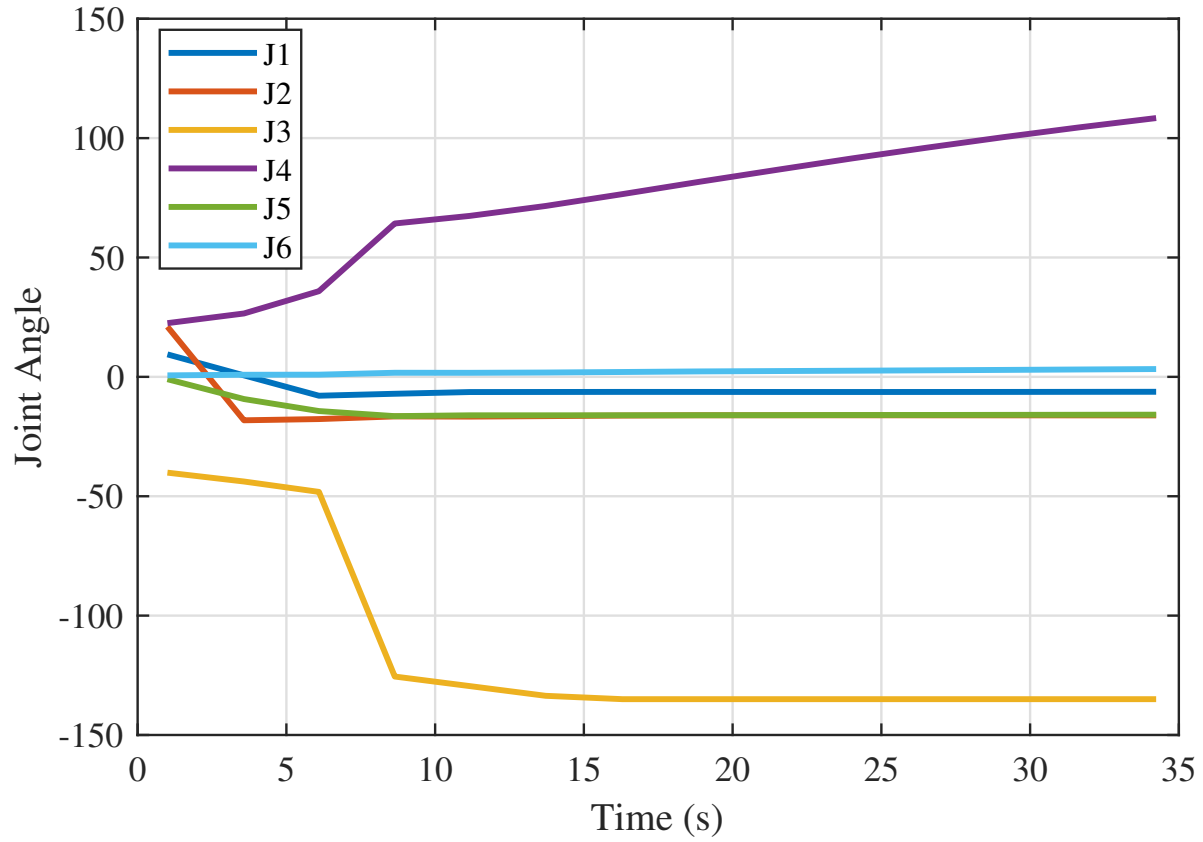


Fig. 12: Case 2 : Elephant robotics Arm joint position vs time.

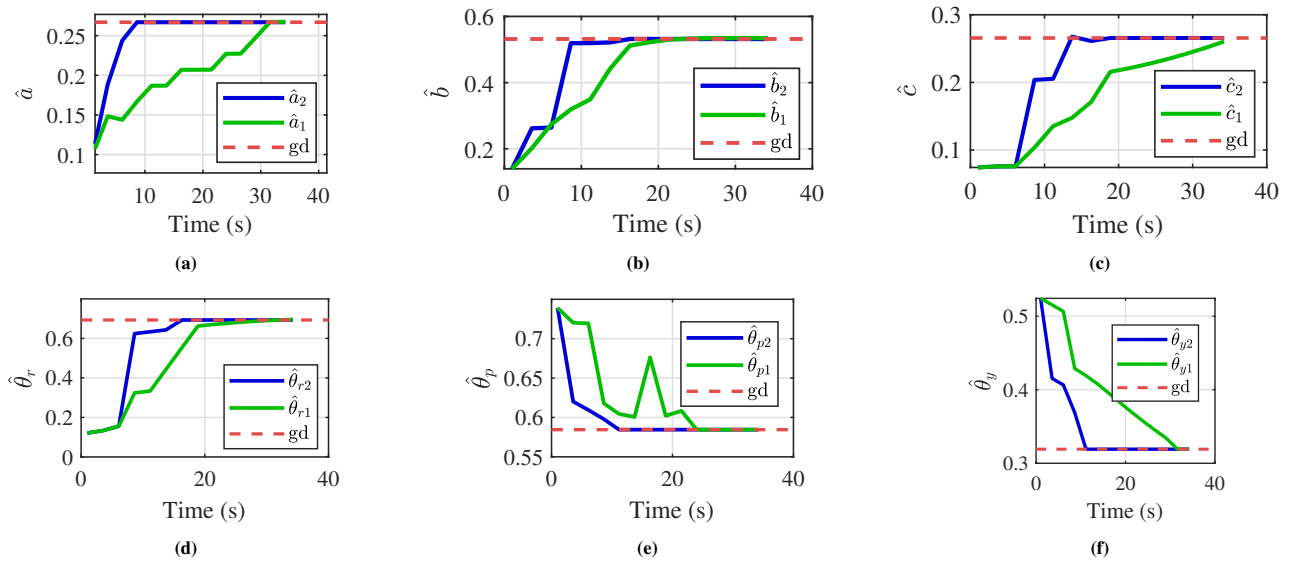


Fig. 13: Case 2: Comparative results of adaptive vs non-adaptive w.r.t the GT for (a) $\hat{a}$, (b) $\hat{b}$, (c) $\hat{c}$, (d) $\hat{\theta}_r$, (e) $\hat{\theta}_p$, and (f) $\hat{\theta}_y$ for the robotic-manipulator.

initial position toward the desired target position. The corresponding Cartesian position errors converge within the predefined error bounds, indicating successful position tracking. The overall 3D RMSE of the end-effector position is 0.12 m.

The joint trajectories shown in Fig. 12 illustrate the motion of all six joints during the reaching task. The joint responses show coordinated motion of the manipulator from the initial configuration toward the desired end-effector position.

The estimated camera parameters of the real Elephant robotic arm are presented in Fig. 13. The results indicate the estimation of the camera parameters for the adaptive case, (denoted by $\hat{a}_1, \hat{b}_1, \hat{c}_1, \hat{\theta}_{r1}, \hat{\theta}_{p1}$ and $\hat{\theta}_{y1}$) vs non-adaptive (denoted by $\hat{a}_2, \hat{b}_2, \hat{c}_2, \hat{\theta}_{r2}, \hat{\theta}_{p2}$ and $\hat{\theta}_{y2}$) over time for the manipulator over time. Overall, the experimental results demonstrate the effectiveness of the proposed approach for Cartesian position tracking and camera parameter estimation on the real Elephant robotic arm. The Demonstration Video of the experiment available at Github

a) **Drone:** : We also conducted a real-time experiment using a custom-built drone. However, the experiment did not perform as expected due to localization issues encountered during real-time operation. We are currently working on addressing these localization challenges and improving the overall reliability of the drone-based real-time experiment. Further experimental results will be reported once the system achieves stable and satisfactory performance. A demonstration video of the real-time drone experiment is available at Github